

MTH 418a: Inference-I

Assignment No. 1: Sampling Distributions

1. For a positive integer ν , let $X \sim t_\nu$. We know that $X \stackrel{d}{=} \frac{Z}{\sqrt{\frac{Y}{\nu}}}$, where $Z \sim N(0, 1)$ and $Y \sim \chi_\nu^2$ are independently distributed. Using the aforementioned representation, find the mean and variance of t_ν . Also show that $t_\nu^2 \sim f_{1, \nu}$.
2. Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$, where $\mu \in (-\infty, \infty)$ and $0 < \sigma < \infty$. Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Find means and variances of r.v.s $S_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$ and $T = \frac{\bar{X}_n}{S_n}$.
3. For positive integers ν_1 and ν_2 , let $X \sim f_{\nu_1, \nu_2}$. We know that $X \stackrel{d}{=} \frac{Y_1/\nu_1}{Y_2/\nu_2}$, where $Y_1 \sim \chi_{\nu_1}^2$ and $Y_2 \sim \chi_{\nu_2}^2$ are independently distributed. Using this representation of f_{ν_1, ν_2} , find its mean and variance. Also show that $T = \frac{1}{f_{\nu_1, \nu_2}} \sim f_{\nu_2, \nu_1}$.
4. Let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ be independent random samples from $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$, respectively, where $\mu_i \in (-\infty, \infty)$ and $0 < \sigma_i < \infty, i = 1, 2$. Let $\bar{X} = \frac{1}{m} \sum_{i=1}^m X_i$, $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$, $S_X^2 = \frac{1}{m} \sum_{i=1}^m (X_i - \bar{X})^2$ and $S_Y^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2$. Find means and variances of following r.v.s:

(a) $T_1 = \frac{\bar{X} - \bar{Y}}{\sqrt{S_X^2 + S_Y^2}}$, when $m = n$ and $\sigma_1^2 = \sigma_2^2$;

(b) $T_2 = \frac{S_X}{S_Y}$.

Also, for $m = n$, find $\lim_{n \rightarrow \infty} P(U_n \leq 1)$, where $U_n = \frac{\sqrt{n}(\bar{X} - \bar{Y} - (\mu_1 - \mu_2))}{\sqrt{\sigma_1^2 + \sigma_2^2}}$.

5. Let $Z_1, \dots, Z_{12}$ be a random sample from $N(0, 1)$. Find the p.d.f.s of the following random variables:

(a) $Y_1 = \frac{Z_1}{\sqrt{\sum_{i=2}^{11} Z_i^2}}$;

(b) $Y_2 = \frac{Z_1 + Z_2}{|Z_1 - Z_2|}$;

(c) $Y_3 = \frac{Z_1 + Z_2}{Z_1 - Z_2}$;

(d) $Y_4 = \frac{Z_1^2 + Z_2^2}{\sum_{i=3}^{11} Z_i^2}$;

(e) $Y_5 = \frac{Z_1^2 + Z_2^2 + Z_3^2}{\sum_{i=1}^{11} Z_i^2}$;

(f) $Y_6 = \sum_{i=1}^9 (Z_i - \bar{Z}_9)^2 + Z_{10}^2$, where $\bar{Z}_9 = \frac{1}{9} \sum_{i=1}^9 Z_i$;

(g) $Y_7 = \frac{3(Z_{11} + Z_{12})}{\sqrt{Y_6}}$;

(h) $Y_8 = \frac{Z_{11}^2 + Z_{12}^2}{2Y_6}$.

6. Let $X_1, X_2, \dots$ be a sequence of r.v.s such that the r.v. X_n follows the Poisson distribution with mean n . Find the values of $\lim_{n \rightarrow \infty} P(\frac{9n}{10} \leq X_n \leq \frac{11n}{10})$, $\lim_{n \rightarrow \infty} P(X_n \leq n + 2\sqrt{n})$ and $\lim_{n \rightarrow \infty} e^{-n} \sum_{j=0}^{[n-\sqrt{n}]} \frac{n^j}{j!}$, where, for any real number x , $[x]$ denotes the largest integer not exceeding x .

Honors Problems

7. Let $X_1, \dots, X_n$ be a random sample from a population having d.f. F and the Lebesgue p.d.f. f . Show that the d.f. of $X_{(r)}$ ($r = 1, \dots, n$) is

$$G_r(y) = \sum_{i=r}^n \binom{n}{i} (F(y))^i (1 - F(y))^{n-i} = \frac{n!}{(r-1)!(n-r)!} \int_0^{F(y)} u^{r-1} (1-u)^{n-r} du.$$

Hence derive the expression for the p.d.f. of $X_{(r)}$ ($r = 1, \dots, n$). Using the above identities, derive the relationship between the d.f.s of binomial and beta distributions.

8. Let $X_1, \dots, X_n$ be a random sample from $Exp(\theta)$ ($0 < \theta < \infty$), and let $X_{(1)}, \dots, X_{(n)}$ be the corresponding order statistics. Define $Y_i = (n - i + 1)(X_{(i)} - X_{(i-1)})$, $i = 1, \dots, n$, where $X_{(0)} = 0$ (Y_i s are called normalized spacings). Show that $Y_1, \dots, Y_n$ are i.i.d. $Exp(\theta)$. Hence, show that $E(X_{(r)}) = \theta \sum_{i=n-r+1}^n \frac{1}{i}$, $Var(X_{(r)}) = \theta^2 \sum_{i=n-r+1}^n \frac{1}{i^2}$, $r = 1, \dots, n$, and $Cov(X_{(r)}, X_{(s)}) = \theta^2 \sum_{i=n-r+1}^n \frac{1}{i^2}$, $1 \leq r < s \leq n$. Further, show that $T = nX_{(1)} \sim Exp(\theta)$.
9. Let X_i s be as defined in Problem 8. Let $U_{(1)}, \dots, U_{(n-1)}$ be the order statistics based on a random sample of size $n - 1$ from $U(0, 1)$. Show that

$$(U_{(1)}, \dots, U_{(n-1)}) \stackrel{d}{=} \left(\frac{X_1}{\sum_{i=1}^n X_i}, \frac{X_1 + X_2}{\sum_{i=1}^n X_i}, \dots, \frac{\sum_{i=1}^{n-1} X_i}{\sum_{i=1}^n X_i} \right).$$

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Assignment No. 1: Sampling Distributions

Problem No. 1

$Z^2 \sim \chi_1^2$ and $Y \sim \chi_0^2$ are independent

$$t_0 = \frac{Z^2/1}{Y/0} = \frac{X_1^2/1}{X_0^2/0} > \text{independent}$$

$$\sim b_{1,0}.$$

Problem No. 2

Recall that if $Y \sim \chi_0^2$, then

$$E(Y^r) = \frac{2^r \Gamma(\frac{0}{2} + r)}{\Gamma(\frac{0}{2})}, \quad r > -\frac{0}{2}.$$

$$\frac{n S_n^2}{\sigma^2} \sim \chi_{n-1}^2 \Rightarrow S_n^2 = \frac{\sigma^2}{n} \chi_{n-1}^2$$

$$E(S_n^2) = \frac{\sigma^2}{n} E(\chi_{n-1}^2) = \frac{n-1}{n} \sigma^2$$

$$\text{Var}(S_n^2) = \frac{\sigma^4}{n^2} \text{Var}(\chi_{n-1}^2) = \frac{2(n-1)\sigma^4}{n^2}$$

Also $\bar{X}_n \sim N(\mu, \frac{\sigma^2}{n})$ and $S_n^2 = \frac{\sigma^2}{n} \chi_{n-1}^2$ are independent. Thus

$$E(T) = E(\bar{X}_n \times \frac{1}{S_n}) = E(\bar{X}_n) \times E(\frac{1}{S_n}) = \mu \times \frac{\sqrt{n}}{\sigma} E(\frac{1}{\chi_{n-1}^2})$$

$$= \frac{\sqrt{n} \mu}{\sigma} \times \frac{2^{-\frac{1}{2}} \Gamma(\frac{n-1}{2} - \frac{1}{2})}{\Gamma(\frac{n-1}{2})} = \sqrt{\frac{n}{2}} \frac{\Gamma(\frac{n-2}{2})}{\Gamma(\frac{n-1}{2})} \frac{\mu}{\sigma}$$

$$E(T^2) = E(\bar{X}_n^2 \times \frac{1}{S_n^2}) = E(\bar{X}_n^2) \times E(\frac{1}{S_n^2}) = E(\bar{X}_n^2) \times \frac{n}{\sigma^2} E(\frac{1}{\chi_{n-1}^2})$$

$$= \frac{n}{\sigma^2} (\mu^2 + \frac{\sigma^2}{n}) \frac{2^{-1} \Gamma(\frac{n-1}{2} - 1)}{\Gamma(\frac{n-1}{2})} = \frac{n}{\sigma^2} (\mu^2 + \frac{\sigma^2}{n}) \times \frac{1}{(n-3)}$$

$$\text{Var}(T) = E(T^2) - (E(T))^2$$

Problem No. 3

$$X = \frac{X_1/\sigma_1}{X_2/\sigma_2} = \frac{X_{\sigma_1}^2/\sigma_1}{X_{\sigma_2}^2/\sigma_2} \rangle \text{ independent}$$

$$T = \frac{1}{b_{\sigma_1, \sigma_2}} = \frac{X_{\sigma_2}^2/\sigma_2}{X_{\sigma_1}^2/\sigma_1} \rangle \text{ independent}$$

$\sim b_{\sigma_1, \sigma_2}$
 $\sim b_{\sigma_2, \sigma_1}$

(this proves the last part of the assertion)

$$X = \frac{\sigma_2}{\sigma_1} \frac{X_{\sigma_1}^2}{X_{\sigma_2}^2} \rangle \text{ independent}$$

$$E(X) = \frac{\sigma_2}{\sigma_1} E(X_{\sigma_1}^2) \times E\left(\frac{1}{X_{\sigma_2}^2}\right) = \frac{\sigma_2}{\sigma_1} \times \sigma_1 \times \frac{2^{-1} \sqrt{\frac{\sigma_2}{2}} - 1}{\sqrt{\sigma_2/2}} = \frac{\sigma_2}{\sigma_2 - 2}, \sigma_2 > 2$$

$$E(X^2) = \frac{\sigma_2^2}{\sigma_1^2} E((X_{\sigma_1}^2)^2) \times E\left(\left(\frac{1}{X_{\sigma_2}^2}\right)^2\right)$$

$$= \frac{\sigma_2^2}{\sigma_1^2} (2\sigma_1 + \sigma_1^2) \times \frac{2^{-2} \sqrt{\frac{\sigma_2}{2}} - 2}{\sqrt{\sigma_2/2}} = \frac{\sigma_2^2 (\sigma_1 + 2)}{\sigma_1 (\sigma_2 - 2) (\sigma_2 - 4)}, \sigma_2 > 4$$

$$V(X) = E(X^2) - (E(X))^2$$

$$= \frac{\sigma_2^2 (\sigma_1 + 2)}{\sigma_1 (\sigma_2 - 2) (\sigma_2 - 4)} - \left(\frac{\sigma_2}{\sigma_2 - 2}\right)^2 = \frac{2\sigma_2^2 (\sigma_1 + \sigma_2 - 2)}{\sigma_1 (\sigma_2 - 2)^2 (\sigma_2 - 4)}, \sigma_2 > 2$$

Problem No. 4

(a) let $m \geq n$ and $\sigma_1^2 = \sigma_2^2 = \sigma^2$. Then

$$\bar{X} - \bar{Y} \sim N\left(\mu_1 - \mu_2, \frac{2\sigma^2}{n}\right)$$

$$\frac{nS_X^2}{\sigma^2} + \frac{nS_Y^2}{\sigma^2} \sim X_{2n-2}^2$$

$$E(T_1) = E(\bar{X} - \bar{Y}) \times E\left(\frac{1}{\sqrt{S_X^2 + S_Y^2}}\right) = (\mu_1 - \mu_2) \times \frac{\sqrt{n}}{\sigma} E\left(\frac{1}{\sqrt{X_{2n-2}^2}}\right)$$

$$= \sqrt{n} \frac{2^{-\frac{1}{2}} \sqrt{\frac{2n-1}{2}} - \frac{1}{2}}{\sqrt{\frac{2n-2}{2}}} \times \frac{\mu_1 - \mu_2}{\sigma} = \sqrt{\frac{n}{2}} \frac{\sqrt{\frac{2n-3}{2}}}{\sqrt{n-1}} \times \frac{\mu_1 - \mu_2}{\sigma}, n \geq 2.$$

$$E(T_1^2) = E((\bar{X} - \bar{Y})^2) \times E\left(\frac{1}{S_X^2 + S_Y^2}\right) = \left(\frac{2\sigma^2}{n} + (\mu_1 - \mu_2)^2\right) \times E\left(\frac{n}{\sigma^2} \times \frac{1}{\chi_{2n-2}^2}\right)$$

$$= n \left(\frac{2}{n} + \left(\frac{\mu_1 - \mu_2}{\sigma}\right)^2\right) \times \frac{2^{-1} \frac{\sqrt{2n-2}-1}{\sqrt{2n-2}}}{\frac{\sqrt{2n-2}}{2}} = \frac{n}{2(n-2)} \left(\frac{2}{n} + \left(\frac{\mu_1 - \mu_2}{\sigma}\right)^2\right)$$

$$\text{Var}(T_1) = E(T_1^2) - (E(T_1))^2$$

(b) $\frac{m S_X^2}{\sigma_1^2} \sim \chi_{m-1}^2$
 $\frac{n S_Y^2}{\sigma_2^2} \sim \chi_{n-1}^2$ } independent $\Rightarrow T_2 = \frac{S_X}{S_Y} = \sqrt{\frac{n}{m}} \frac{\sigma_1}{\sigma_2} \frac{\sqrt{\chi_{m-1}^2}}{\sqrt{\chi_{n-1}^2}}$ indept.

$$E(T_2) = \sqrt{\frac{n}{m}} \frac{\sigma_1}{\sigma_2} E(\sqrt{\chi_{m-1}^2}) \times E\left(\frac{1}{\sqrt{\chi_{n-1}^2}}\right)$$

$$= \sqrt{\frac{n}{m}} \frac{\sigma_1}{\sigma_2} \frac{2^{1/2} \frac{\sqrt{m-1} + \frac{1}{2}}{\sqrt{m-2}}}{\frac{\sqrt{m-1}}{2}} \times \frac{2^{-1/2} \frac{\sqrt{n-1} - \frac{1}{2}}{\sqrt{n-2}}}{\frac{\sqrt{n-1}}{2}}$$

$$= \sqrt{\frac{n}{m}} \frac{\frac{\sqrt{m-2}}{\sqrt{m-1}} \frac{\sqrt{n-2}}{\sqrt{n-1}}}{\frac{\sqrt{m-1}}{\sqrt{n-1}}} \frac{\sigma_1}{\sigma_2}$$

$$E(T_2^2) = \frac{n}{m} \frac{\sigma_1^2}{\sigma_2^2} E(\chi_{m-1}^2) \times E\left(\frac{1}{\chi_{n-1}^2}\right) = \frac{n}{m} \frac{\sigma_1^2}{\sigma_2^2} \times (m-1) \times \frac{2^{-1} \frac{\sqrt{n-1}-1}{\sqrt{n-2}}}{\frac{\sqrt{n-1}}{2}}$$

$$= \frac{n(n-1)}{m(n-3)} \frac{\sigma_1^2}{\sigma_2^2}; \text{Var}(T_2) = E(T_2^2) - (E(T_2))^2$$

Let $T_i = X_i - Y_i$, $i=1, 2, \dots, n$. Then $T_1, T_2, \dots, T_n$ are iid $N(\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2)$. $\bar{T} = \frac{1}{n} \sum_{i=1}^n T_i = \frac{1}{n} \sum_{i=1}^n (X_i - Y_i) = \bar{X} - \bar{Y}$.

By the CLT, $\frac{\sqrt{n}(\bar{T} - (\mu_1 - \mu_2))}{\sqrt{\sigma_1^2 + \sigma_2^2}} \xrightarrow{d} Z \sim N(0,1)$, as $n \rightarrow \infty$

i.e., $U_n = \frac{\sqrt{n}(\bar{X} - \bar{Y} - (\mu_1 - \mu_2))}{\sqrt{\sigma_1^2 + \sigma_2^2}} \xrightarrow{d} Z \sim N(0,1)$, as $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} P(U_n \leq 1) = P(Z \leq 1) = \Phi(1) = 0.8413$$

Problem 410.5

Recall that p.d.f. of χ^2_ν distribution is

$$b_1(x) = \frac{1}{2^{\nu/2} \Gamma(\nu/2)} e^{-x/2} x^{\nu/2-1}, \quad x > 0,$$

the p.d.f. of t_ν distribution is

$$b_2(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi} \Gamma(\frac{\nu}{2})} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \quad -\infty < x < \infty$$

and the p.d.f. of b_{ν_1, ν_2} distribution is

$$b_3(x) = \frac{1}{B(\frac{\nu_1}{2}, \frac{\nu_2}{2})} \left(\frac{\nu_1}{\nu_2}\right)^{\frac{\nu_1}{2}} b^{\frac{\nu_1}{2}-1} \left(1 + \frac{\nu_1}{\nu_2} b\right)^{-\frac{\nu_1+\nu_2}{2}}, \quad b > 0.$$

(a) $Z_1 \sim N(0, 1)$ and $\sum_{i=2}^{10} Z_i^2 \sim \chi^2_{10}$ are independent

$$\Rightarrow T = \frac{Z_1}{\sqrt{\frac{\sum_{i=2}^{10} Z_i^2}{10}}} \sim t_{10}, \quad \text{i.e. } T = \sqrt{10} \gamma_1 \sim t_{10}$$

The p.d.f. of T is

$$b_T(t) = \frac{\Gamma(\frac{11}{2})}{\sqrt{10\pi} \Gamma(5)} \left(1 + \frac{t^2}{10}\right)^{-\frac{11}{2}}, \quad -\infty < t < \infty.$$

The p.d.f. of γ_1 is

$$b_{\gamma_1}(\gamma) = \frac{\Gamma(\frac{11}{2})}{\sqrt{10\pi} \Gamma(5)} (1 + \gamma^2)^{-\frac{11}{2}} \times \sqrt{10}, \quad -\infty < \gamma < \infty$$

$$= \frac{\Gamma(\frac{11}{2})}{24\sqrt{\pi}} \frac{1}{(1 + \gamma^2)^{\frac{11}{2}}}, \quad -\infty < \gamma < \infty.$$

(b) $\begin{pmatrix} Z_1 + Z_2 \\ Z_1 - Z_2 \end{pmatrix} \sim N_2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \right)$

Thus $Z_1 + Z_2$ and $Z_1 - Z_2$ are iid $N(0, 2)$

$\frac{Z_1 + Z_2}{\sqrt{2}} \sim N(0, 1)$ and $\left(\frac{Z_1 - Z_2}{\sqrt{2}}\right)^2 \sim \chi^2_1$ are independent

$$\frac{(Z_1 + Z_2)/\sqrt{2}}{\sqrt{\left(\frac{Z_1 - Z_2}{\sqrt{2}}\right)^2}} \sim t_1, \quad \text{i.e. } \gamma_2 = \frac{Z_1 + Z_2}{Z_1 - Z_2} \sim t_1$$

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The p.d.f. of γ_2 is

$$b_{\gamma_2}(y) = \frac{1}{\pi} \cdot \frac{1}{1+y^2}, \quad -\infty < y < \infty$$

(c) $\frac{z_1+z_2}{\sqrt{2}}$ and $\frac{z_1-z_2}{\sqrt{2}}$ are iid $N(0,1)$

$$\Rightarrow \frac{(z_1+z_2)/\sqrt{2}}{(z_1-z_2)/\sqrt{2}} \sim \text{Cauchy}(0,1) \equiv t_1$$

$$\Rightarrow \gamma_3 = \frac{z_1+z_2}{z_1-z_2} \sim t_1$$

The p.d.f. of γ_3 is

$$b_{\gamma_3}(y) = \frac{1}{\pi} \cdot \frac{1}{1+y^2}, \quad -\infty < y < \infty$$

(d) $z_1^2+z_2^2 \sim X_2^2$ and $\sum_{i=3}^{11} z_i^2 \sim X_9^2$ are independent

$$\Rightarrow F = \frac{(z_1^2+z_2^2)/2}{\sum_{i=3}^{11} z_i^2/9} \sim F_{2,9} \quad \text{i.e. } F = \frac{9}{2} \gamma_4 \sim b_{2,9}$$

The p.d.f. of F is

$$g_F(b) = \frac{1}{B(1, \frac{9}{2})} \left(\frac{2}{9}\right)^1 b^0 (1 + \frac{2}{9}b)^{-\frac{11}{2}}, \quad b > 0$$

The p.d.f. of γ_4 is

$$b_{\gamma_4}(y) = \frac{\frac{\Gamma(\frac{11}{2})}{\Gamma(\frac{1}{2})}}{\Gamma(\frac{9}{2})} \frac{1}{(1+y)^{\frac{11}{2}}} = \frac{9}{2} \cdot \frac{1}{(1+y)^{\frac{11}{2}}}, \quad y > 0$$

(e) Recall that if $X \sim \text{Gamma}(\theta, m)$ and $Y \sim \text{Gamma}(\theta, n)$ are independent, then $\frac{X}{X+Y} \sim \text{Beta}(m, n)$

$$T_1 = z_1^2 + z_2^2 + z_3^2 \sim \text{Gamma}(\frac{1}{2}, \frac{3}{2}) \equiv X_3^2$$

$$T_2 = \sum_{i=4}^{11} z_i^2 \sim \text{Gamma}(\frac{1}{2}, 4) \equiv X_4^2$$

> Independent

$$\gamma_5 = \frac{T_1}{T_1+T_2} \sim \text{Beta}(\frac{3}{2}, 4)$$

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The p.d.f. of γ_5 is

$$b_{\gamma_5}(\gamma) = \frac{1}{B(\frac{3}{2}, 4)} \gamma^{\frac{3}{2}} (1+\gamma)^3, \quad 0 < \gamma < 1.$$

(b) $\sum_{i=1}^9 (z_i - \bar{z}_9)^2 \sim \chi_8^2$ and $z_{10}^2 \sim \chi_1^2$ are independent

$$\gamma_6 = \sum_{i=1}^9 (z_i - \bar{z}_9)^2 + z_{10}^2 \sim \chi_9^2$$

The p.d.f. of γ_6 is

$$b_{\gamma_6}(\gamma) = \frac{1}{2^{9/2} \Gamma(\frac{9}{2})} e^{-\frac{\gamma}{2}} \gamma^{\frac{9}{2}-1}, \quad \gamma > 0$$

(g) $\frac{z_{11} + z_{12}}{\sqrt{2}} \sim N(0, 1)$ and $\gamma_6 \sim \chi_9^2$ are independent

$$\Rightarrow T = \frac{(z_{11} + z_{12})/\sqrt{2}}{\sqrt{\gamma_6/9}} \sim t_9 \Rightarrow T = \frac{\gamma_7}{\sqrt{2}} \sim t_9$$

The p.d.f. of T is

$$b_{T|t=1} = \frac{\sqrt{5}}{\sqrt{9\pi} \Gamma(\frac{5}{2})} \left(1 + \frac{t^2}{9}\right)^{-5}, \quad -\infty < t < \infty$$

The p.d.f. of $\gamma_7 = \sqrt{2}T$ is

$$b_{\gamma_7}(\gamma) = \frac{8}{\sqrt{2\pi} \Gamma(\frac{5}{2})} \frac{1}{(1 + \frac{\gamma^2}{18})^5}, \quad -\infty < \gamma < \infty.$$

(h) $z_{11}^2 + z_{12}^2 \sim \chi_2^2$ and $\gamma_6 \sim \chi_9^2$ are independent

$$\Rightarrow F = \frac{(z_{11}^2 + z_{12}^2)/2}{\gamma_6/9} \sim b_{2,9}, \quad \text{i.e. } F = 9\gamma_8 \sim b_{2,9}$$

The p.d.f. of F is

$$g_F(b) = \frac{1}{B(1, \frac{9}{2})} \left(\frac{2}{9}\right) \left(1 + \frac{2}{9}b\right)^{-\frac{11}{2}}, \quad b > 0$$

The p.d.f. of γ_8 is

$$b_{\gamma_8}(\gamma) = \frac{9}{(1+2\gamma)^{-\frac{11}{2}}}, \quad \gamma > 0.$$

(6A)

Problem No. 6

We can write as

$$X_n \stackrel{d}{=} \sum_{i=1}^n Y_i$$

Where $Y_1, Y_2, \dots, Y_n$ are iid $\text{Poisson}(1)$

$$\mu = E(Y_i) = 1, \quad \sigma^2 = \text{Var}(Y_i) = 1$$

By the CLT, for $\bar{Y}_n = \frac{1}{n} \sum_{i=1}^n Y_i$

$$Z_n = \frac{\sqrt{n}(\bar{Y}_n - \mu)}{\sigma} \xrightarrow{d} Z \sim N(0, 1)$$

$$\text{i.e. } Z_n = \sqrt{n} \left(\frac{X_n}{n} - 1 \right) \xrightarrow{d} Z \sim N(0, 1)$$

$$\text{or } Z_n = \frac{X_n - n}{\sqrt{n}} \xrightarrow{d} Z \sim N(0, 1)$$

$$\lim_{n \rightarrow \infty} P\left(\frac{9n}{10} \leq X_n \leq \frac{11n}{10}\right) = \lim_{n \rightarrow \infty} P\left(-\frac{\sqrt{n}}{10} \leq Z_n \leq \frac{\sqrt{n}}{10}\right)$$

$$= \lim_{n \rightarrow \infty} \left[P\left(Z_n \leq \frac{\sqrt{n}}{10}\right) - \lim_{n \rightarrow \infty} P\left(Z_n < -\frac{\sqrt{n}}{10}\right) \right]$$

$$= \Phi(+\infty) - \Phi(-\infty) = 1$$

$$\lim_{n \rightarrow \infty} P(X_n \leq n + 2\sqrt{n}) = \lim_{n \rightarrow \infty} P(Z_n \leq 2) = \Phi(2) = 0.9772$$

$$\lim_{n \rightarrow \infty} e^{-n} \sum_{j=0}^{\lfloor n - \sqrt{n} \rfloor} \frac{n^j}{j!} = \lim_{n \rightarrow \infty} P(X_n \leq \lfloor n - \sqrt{n} \rfloor)$$

$$= \lim_{n \rightarrow \infty} P\left(Z_n \leq \frac{\lfloor n - \sqrt{n} \rfloor - n}{\sqrt{n}}\right)$$

$$\frac{\lfloor n - \sqrt{n} \rfloor - 1 - n}{\sqrt{n}} \leq \frac{\lfloor n - \sqrt{n} \rfloor - n}{\sqrt{n}} \leq \frac{n - \sqrt{n} - n}{\sqrt{n}} \quad [21 \leq [x] \leq x]$$

$$-1 - \frac{1}{\sqrt{n}} \leq \frac{\lfloor n - \sqrt{n} \rfloor - n}{\sqrt{n}} \leq -1 \Rightarrow \lim_{n \rightarrow \infty} \frac{\lfloor n - \sqrt{n} \rfloor - n}{\sqrt{n}} = -1$$

$$\text{Thus } \lim_{n \rightarrow \infty} e^{-n} \sum_{j=0}^{\lfloor n - \sqrt{n} \rfloor} \frac{n^j}{j!} = P(Z \leq -1) = \Phi(-1) = 0.1587$$

Problem No. 7

Let $y \in (-\infty, \infty)$ be fixed. Then

$$\begin{aligned} G_r(y) &= P(X_{(r)} \leq y) = P(\text{at least } r \text{ of } x_1, \dots, x_n \text{ are } \leq y) \\ &= \sum_{i=r}^n P(\text{exactly } i \text{ of } x_1, \dots, x_n \text{ are } \leq y) \\ &= \sum_{i=r}^n P(N=i), \end{aligned}$$

Where $N = \#$ of x_i 's among $x_1, \dots, x_n \leq y$
 $\sim \text{Bin}(n, F(y))$ (independent trials) where prob. of success in i th trial $= P(x_i \leq y) = F(y)$

$$G_r(y) = \sum_{i=r}^n \binom{n}{i} (F(y))^i (1-F(y))^{n-i} \dots \text{ (I)}$$

Define $U_i = F(x_i)$, $i=1, \dots, n$. Then $U_1, \dots, U_n$ are iid $U(0,1)$.
 $U_{(r)} = r$ th smallest of $U_1, \dots, U_n = F(X_{(r)})$, $r=1, \dots, n$ (as $F(\cdot) \uparrow$)

Then

$$\begin{aligned} G_r(y) &= P(X_{(r)} \leq y) = P(F(X_{(r)}) \leq F(y)) = P(U_{(r)} \leq F(y)) \\ &= \int_0^{F(y)} \frac{1}{(r-1)! (n-r)!} u^{r-1} (1-u)^{n-r} du \quad \left(\begin{array}{l} \text{for } U(0,1), \text{ d.f. } g(y) = g'(y) \\ \text{and p.d.f. } g(y) = 1, 0 \leq y \leq 1 \end{array} \right) \end{aligned}$$

... (II)

Taking $F(y) = p$, in (I) and (II) we get

$$\frac{1}{B(r, n-r)} \int_0^p u^{r-1} (1-u)^{n-r} du = \sum_{i=r}^n \binom{n}{i} p^i (1-p)^{n-i}$$

Problem No. 8

The joint p.d.f. of $(x_{(1)}, \dots, x_{(n)})$ is

$$f_{x_{(1)}, \dots, x_{(n)}}(\lambda_1, \dots, \lambda_n) = \frac{1}{\theta^n} e^{-\frac{1}{\theta} \sum_{i=1}^n \lambda_i}, \quad \lambda_i > 0, \quad i=1, \dots, n$$

Making the transformation $y_i = (n-i+1)(\lambda_i - \lambda_{i-1})$, $i=1, \dots, n$, we have

$$y_1 = n\lambda_1, \quad y_2 = (n-1)(\lambda_2 - \lambda_1), \quad y_3 = (n-2)(\lambda_3 - \lambda_2), \dots, \quad y_i = (n-i+1)(\lambda_i - \lambda_{i-1}),$$

$$\dots, \quad y_{n-1} = 2(\lambda_{n-1} - \lambda_{n-2}), \quad y_n = \lambda_n - \lambda_{n-1}$$

$$\Rightarrow \lambda_1 = \frac{y_1}{n}, \quad \lambda_2 = \frac{y_1}{n} + \frac{y_2}{n-1}, \quad \lambda_3 = \frac{y_1}{n} + \frac{y_2}{n-1} + \frac{y_3}{n-2}, \dots, \quad \lambda_i = \frac{y_1}{n} + \frac{y_2}{n-1} + \frac{y_3}{n-2} + \dots + \frac{y_i}{n-i+1}$$

$$\dots, \quad \lambda_{n-1} = \frac{y_1}{n} + \frac{y_2}{n-1} + \frac{y_3}{n-2} + \dots + \frac{y_{n-1}}{2}, \quad \lambda_n = \frac{y_1}{n} + \frac{y_2}{n-1} + \frac{y_3}{n-2} + \dots + \frac{y_{n-1}}{2} + y_n$$

$$J = \frac{1}{n!}, \quad x_i > 0, \quad i=1, \dots, n \Rightarrow y_i > 0, \quad i=1, \dots, n, \quad \text{Also } \sum_{i=1}^n \lambda_i = \sum_{i=1}^n y_i$$

Then the joint p.d.f. of $(y_1, \dots, y_n)$ is

$$f_{y_1, \dots, y_n}(y_1, \dots, y_n) = \frac{1}{\theta^n} e^{-\frac{1}{\theta} \sum_{i=1}^n y_i}, \quad y_i > 0, \quad i=1, \dots, n$$

$$= \prod_{i=1}^n \frac{1}{\theta} e^{-y_i/\theta}, \quad y_i > 0, \quad i=1, \dots, n$$

$\Rightarrow y_1, y_2, \dots, y_n$ are iid $\text{Exp}(\theta)$ ($E(y_i) = \theta, \quad V(y_i) = \theta^2$)

$$X(r) = \sum_{i=1}^r \frac{y_i}{n-i+1}$$

$$\Rightarrow E(X(r)) = \sum_{i=1}^r \frac{1}{n-i+1} E(y_i) = \theta \sum_{i=1}^r \frac{1}{n-i+1} = \theta \sum_{i=n-r+1}^n \frac{1}{i}$$

$$V(X(r)) = \sum_{i=1}^r \left(\frac{1}{n-i+1} \right)^2 V(y_i) = \theta^2 \sum_{i=1}^r \frac{1}{(n-i+1)^2} = \theta^2 \sum_{i=n-r+1}^n \frac{1}{i^2}$$

For $1 \leq r < n \leq n$,

$$\text{Cov}(X(r), X(n)) = \text{Cov}\left(\sum_{i=1}^r \frac{y_i}{n-i+1}, \sum_{i=1}^n \frac{y_i}{n-i+1}\right)$$

$$= \text{Cov}\left(\sum_{i=1}^r \frac{y_i}{n-i+1}, \sum_{i=1}^r \frac{y_i}{n-i+1} + \sum_{i=r+1}^n \frac{y_i}{n-i+1}\right)$$

$$= \text{Cov}\left(\sum_{i=1}^r \frac{y_i}{n-i+1}, \sum_{i=1}^r \frac{y_i}{n-i+1}\right) = \text{Var}\left(\sum_{i=1}^r \frac{y_i}{n-i+1}\right)$$

$$= \theta^2 \sum_{i=1}^r \frac{1}{i^2}$$

(clearly) $T = n X(n) = y_1 \sim \text{Exp}(\theta)$

Problem 44.9

The joint p.d.b. of $(U_1, \dots, U_{n-1})$ is

$$f_{U_1, \dots, U_{n-1}}(u_1, \dots, u_{n-1}) = \frac{n-1}{1}, \quad 0 < u_1 < u_2 < \dots < u_{n-1} < 1. \quad (I)$$

$$\text{Let } \left(\frac{x_1}{\sum_{i=1}^n x_i}, \frac{x_1+x_2}{\sum_{i=1}^n x_i}, \dots, \frac{\sum_{i=1}^{n-1} x_i}{\sum_{i=1}^n x_i} \right) = (T_1, \dots, T_{n-1}).$$

It suffices to show that the joint p.d.b. of $(T_1, \dots, T_{n-1})$ is the same as (I).

The joint p.d.b. of $(x_1, \dots, x_n)$ is

$$f_{x_1, \dots, x_n}(x_1, \dots, x_n) = \frac{1}{\theta^n} e^{-\frac{1}{\theta} \sum_{i=1}^n x_i}, \quad x_i > 0, \quad i=1, \dots, n$$

Define $T_n = \sum_{i=1}^n x_i$. Making the transformation

$$t_1 = \frac{x_1}{\sum_{i=1}^n x_i}, \quad t_2 = \frac{x_1+x_2}{\sum_{i=1}^n x_i}, \quad t_3 = \frac{x_1+x_2+x_3}{\sum_{i=1}^n x_i}, \dots, \quad t_{n-1} = \frac{\sum_{i=1}^{n-1} x_i}{\sum_{i=1}^n x_i}, \quad t_n = \frac{\sum_{i=1}^n x_i}{\sum_{i=1}^n x_i}$$

we have

$$x_1 = t_1 t_n, \quad x_2 = (t_2 - t_1) t_n, \quad x_3 = (t_3 - t_2) t_n, \dots, \quad x_{n-1} = (t_{n-1} - t_{n-2}) t_n, \quad x_n = (1 - t_{n-1}) t_n$$

$$J = \frac{n-1}{t_n}$$

$$x_i > 0, \quad i=1, \dots, n \Rightarrow 0 < t_1 < t_2 < \dots < t_{n-1} < t_n < 1, \quad t_n > 0$$

The joint p.d.b. of $(T_1, \dots, T_{n-1}, T_n)$ is

$$f_{T_1, \dots, T_{n-1}, T_n}(t_1, \dots, t_{n-1}, t_n) = \frac{1}{\theta^n} e^{-\frac{t_n}{\theta}} \frac{n-1}{t_n}, \quad 0 < t_1 < t_2 < \dots < t_{n-1} < t_n < 1, \quad t_n > 0$$

The joint p.d.b. of $(T_1, \dots, T_{n-1})$ is

$$f_{T_1, \dots, T_{n-1}}(t_1, \dots, t_{n-1}) = \int_0^1 \frac{1}{\theta^n} e^{-\frac{t_n}{\theta}} \frac{n-1}{t_n} dt_n, \quad 0 < t_1 < t_2 < \dots < t_{n-1} < 1$$

$$= \frac{n-1}{1}, \quad 0 < t_1 < \dots < t_{n-1} < 1,$$

which matches with (I).

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